

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

Nov – Dec 2016

Max. Marks: 100

Class: SYBTech

Name of the Course: Data Structures

Course Code: UCEC303

Duration: 3 hrs

Semester: III

Branch: COMP

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.	or C program	Max. Marks
Q 1 (a)	Write pseudo code for the following functions in stack i) <i>push(element)</i> ii) <i>pop()</i> iii) <i>display_top()</i> iv) <i>isEmpty()</i> v) <i>isFull()</i>	08
Q 1 (b)	Write a program for implementation of Circular Queue. Consider all basic methods of queue. OR Write a program for implementation of Double Ended Queue. Consider all basic methods of queue.	12
Q2 (a)	Write a Program for addition and subtraction of polynomials using appropriate data structure.	15
Q2 (b)	Explain with a suitable example use of stack in recursion	05
Q3 (a)	Construct AVL tree for the elements given below in sequence 10, 20, 30, 25, 40, 50, 35, 33 Perform following operations on the constructed AVL tree i) Insert 37, 60 and 38 ii) Delete 20 and 35 Show AVL tree after completion of each and every task.	10
Q3 (b)	Write algorithm for DFS graph traversal and explain with suitable example OR Write algorithm for BFS graph traversal and explain with suitable example	10
Q4	WAP for implementation of BST . Provide an option to give sorted elements in ascending order for tree formation function. Use function named <i>insertion()</i> for sorting the elements, which uses insertion sort. OR WAP for implementation of BST . Provide an option to give sorted elements in ascending order for tree formation function. Use function named <i>selection()</i> for sorting the elements, which uses selection sort.	20
Q 5 a	Explain Hashing and collision. Explain different collision resolution methods in hashing with a suitable example.	10
Q5 b	Write algorithm for priority queue using heap. Demonstrate the same with a suitable example	10